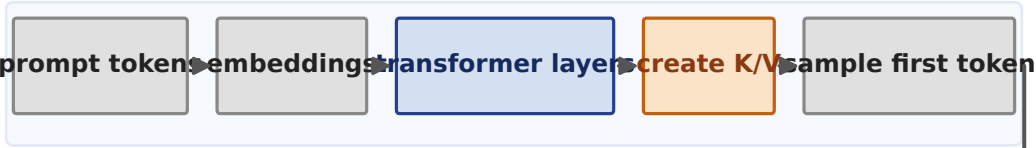


PREFILL

process the whole prompt, populate the KV cache, emit the first token



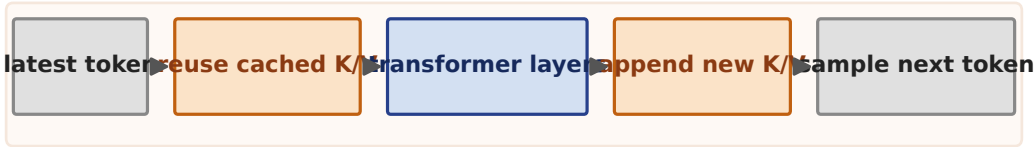
one-shot burst — parallel across prompt positions; measured by TTFT



**KV cache: created in prefill,
reused (not recomputed) in decode**

DECODE

generate one token at a time, reusing the cached K/V



sequential loop — one token per step, reuses + appends K/V; measured by TPOT / ITL

resource regime

a property of the model $\times$ workload $\times$ kernel $\times$ hardware, not a fixed rule

Prefill: typically higher arithmetic intensity | Decode: often bandwidth-sensitive at low batch

visual grammar:

data / tokens model / compute KV state (persistent)